

College of Industrial Technology
King Mongkut's University of Technology North Bangkok



Midterm Examination of Semester 1

Year: 2015

Subject: 394173 Mathematics III

Section: 15-16

Date: 13 October 2015

Time: 13.00-15.00

Name _____ ID _____ Class _____

Instructions

1. Cheating will result in failure of all classes registered for the current semester.
Students who are caught cheating will also be denied registering for the following semester.
2. **No** documents are allowed to be taken out of the examination room.
3. Textbooks and dictionary are **Not** allowed.
4. Calculators are **Not** allowed in the examination.
5. This is a closed book examination.
6. Electronic communication device are **Not** allow in the examination room.
7. The examination has 9 pages (including this page), 8 questions and 1 part.
The total score is 35 points.
8. Write all your answers on this examination sheet.

No.	score
1.	
2.	
3.	
4.	
5.	
6.	
7.	
8.	
Total	

1. Assume that $z_1 = \frac{5-12i^{85}}{26i^{15}}$ and $z_2 = \frac{i}{4i^{62} + 6i^{2015}}$. Evaluate $\left| \frac{\overline{z_2}(z_1)^2}{-z_1} \right|$.
(5 points)

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2. Find the sum of absolute value of solutions to

$$z^5 - 2z^4 + 4iz^4 - 2z^3 - 8iz^3 + z^2 - 8iz^2 + 4iz = 0.$$

(4 points)

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3. Let z_0, z_1 and z_2 be 3rd roots of $-1 + \sqrt{3}i$, find $\left| \frac{z_1 \cdot z_2}{z_0^2} \right|$. (5 points)

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4. Let $2A^T = \begin{bmatrix} -4 & -8 & -2 \\ 2 & -4 & 0 \\ 2 & 0 & -4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ -2 & -1 \\ 2 & 3 \end{bmatrix} \cdot \begin{bmatrix} -1 & -2 & -3 \\ 1 & 2 & 1 \end{bmatrix}$.

Find the skew-symmetric matrix of $A - B$.

(4 points)

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5. Given that $\begin{vmatrix} -2 & 1 & 1 \\ -2 & -1 & 2 \\ 1 & 5 & 3 \end{vmatrix} = \begin{vmatrix} x & 0 & -x \\ 3 & 0 & 2 \\ -4 & 1 & 1 \end{vmatrix}$ and $B = \begin{bmatrix} 2 & 0 & x & 0 \\ 1 & 0 & 2 & -1 \\ -x & 0 & -2 & 3 \\ -1 & 2 & 4 & -x \end{bmatrix}$.

Determine the value of $\det\left(\frac{3B^2}{B^T}\right)$.

(4 points)

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6. Evaluate A^{-1} , where $A \cdot \begin{bmatrix} 3 & 0 & 1 \\ 5 & -1 & 2 \\ -4 & 1 & -1 \end{bmatrix} = \begin{bmatrix} 4 & -2 & -3 \\ 0 & 3 & 2 \\ -1 & 2 & 3 \end{bmatrix}$.

(5 points)

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7. Solve x_3 , using Cramer's rule:

(4 points)

$$x_1 + x_2 = 0$$

$$x_2 + x_3 - 2x_4 = 1$$

$$x_1 + 2x_3 + x_4 = 0$$

$$x_1 + x_2 + x_4 = 0.$$

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8. Solve the system of equations using Gauss–Jordan elimination:

(4 points)

$$\begin{array}{rrcrcl} x_1 & + & 2x_2 & - & 3x_3 & + & 5x_4 & = & 0 \\ & & 3x_2 & - & x_3 & + & 4x_4 & = & 2 \\ 2x_1 & + & 2x_2 & + & 4x_3 & - & 2x_4 & = & 8 \\ x_1 & + & x_2 & - & 5x_3 & + & 6x_4 & = & -3 \end{array}$$

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